THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bag has 5 red, 5 black and 5 white balls. The probability of drawing a ball that is not black is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{2}$

2. If $P(E) = 0.6$, then $P(\text{not } E) \times 100$ equals: 1

(a) 39 **(b)** 40 **(c)** 41 **(d)** 42

3. A fair die is thrown once. The probability of getting a number not divisible by 3 is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

4. The set of all possible outcomes of an experiment is its: 1

(a) sample space **(b)** probability **(c)** frequency **(d)** event

5. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 18, 12, 20, 15, 19, 16. The experimental probability of getting 5 is: 1

(a) $\frac{19}{100}$ **(b)** $\frac{19}{101}$ **(c)** $\frac{81}{100}$ **(d)** $\frac{1}{5}$

6. In a survey of 600 families, 210 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{7}{20}$ **(c)** $\frac{13}{20}$ **(d)** $\frac{1}{3}$

7. A letter is chosen at random from the word CHANDIGARH. The probability that it is a vowel is: 1

(a) $\frac{3}{10}$ **(b)** $\frac{7}{10}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{3}{11}$

8. Three coins are tossed together. The probability of getting exactly two tails is: 1

(a) $\frac{5}{8}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{3}{8}$

9. A bag has 7 red, 9 black and 4 white balls. The probability of drawing a ball that is not black is: 1

(a) $\frac{11}{21}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{11}{20}$ **(d)** $\frac{9}{20}$

10. If $P(E) = 0.7$, then $P(\text{not } E) \times 100$ equals: 1

(a) 31 **(b)** 30 **(c)** 32 **(d)** 29

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Three coins are tossed together. Find the probability of getting no tail. 2

12. A letter is chosen at random from the word EDUCATION. Find the probability that it is a vowel. 2

13. A letter is chosen at random from the word EXAMINATION. Find the probability that it is a consonant. 2

14. Three coins are tossed together. Find the probability of getting exactly two tails. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 3 red, 8 black and 4 white balls. A ball is drawn at random. Find the probability that it is — 3
(i) red,
(ii) not red,
(iii) black or white.

16. In a survey of 180 students, 60 take the bus, 40 cycle and the rest walk. Find the probability that a randomly chosen student — 3
(i) walks,
(ii) does not take the bus.

17. Two coins are tossed 300 times: two heads occurred 72 times, one head 150 times and no head 78 times. Find the experimental probability of each of the three events. — 3

OR

A bag contains 2 red, 6 black and 7 white balls. A ball is drawn at random. Find the probability that it is — 3
(i) red,
(ii) not red,
(iii) black or white.

18. In a survey of 400 students, 150 take the bus, 100 cycle and the rest walk. Find the probability that a randomly chosen student — 3
(i) walks,
(ii) does not take the bus.

OR

Two coins are tossed 50 times: two heads occurred 12 times, one head 27 times and no head 11 times. Find the experimental probability of each of the three events. — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A die was thrown 120 times; the number of times $1, \ldots, 6$ appeared was 25, 18, 16, 22, 19, 20. Find the experimental probability of — 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A letter is chosen at random from the word CHANDIGARH. Find the probability that it is — 5
(i) a vowel,
(ii) the letter H,
(iii) a letter from the first half of the alphabet (A to M).

20. A die was thrown 120 times; the number of times $1, \ldots, 6$ appeared was 20, 24, 18, 17, 22, 19. Find the experimental probability of — 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A letter is chosen at random from the word RAJASTHAN. Find the probability that it is — 5
(i) a vowel,
(ii) the letter A,
(iii) a letter from the first half of the alphabet (A to M).

END OF QUESTION PAPER — Total: 40 Marks

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